

# PHOEBE CHEN

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## EDUCATION

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### UNIVERSITY OF CALIFORNIA, LOS ANGELES

Expected Jun 2026

B.S. in **Statistics and Data Science** | Minor in **Data Science Engineering**

Relevant Coursework: Machine Learning, Neural Networks & Deep Learning, Experimental Design, Causal Inference, Regression & Data Mining, Computational Statistics, Database Management Systems, Linear Models, Bayesian Statistics

### MOORPARK COMMUNITY COLLEGE

Aug 2022 – May 2024

Honors Program

## PROJECTS

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### Semantic Matching & Foundation-Model Evaluation

Jun 2026

- Evaluated bi-encoder, cross-encoder, and LoRA-fine-tuned language models for semantic matching and mathematical reasoning using accuracy, F1, latency, and error analysis.

### Voter Turnout Analysis — [GitHub](#)

Jun 2026

- Built and compared XGBoost turnout models on 33,315 voter records and 900 candidate variables; evaluated ROC-AUC and non-voter recall to select an interpretable 18-feature model achieving 0.803 ROC-AUC and 89.9% recall for 2027 municipal outreach.

### EMG-Based Keystroke Decoding with Neural Networks — [GitHub](#)

Mar 2026

- Designed and evaluated a CNN-BiGRU sequence model in PyTorch, achieving approximately 16% character error rate through iterative architecture experiments and hyperparameter optimization.

### Causal Analysis of Obesity and ICU Mortality

Mar 2026

- Evaluated the effect of obesity on ICU mortality using logistic regression and inverse probability weighting; assessed robustness across alternative specifications and investigated treatment-effect heterogeneity using causal forests.

### LAPD Crime Data Analysis — [GitHub](#)

Mar 2026

- Performed large-scale EDA and statistical modeling on 877K+ LAPD crime records in R, identifying temporal, spatial, and post-COVID patterns and communicating results through reproducible ggplot2 and Quarto analyses.

### Heart Failure Mortality Prediction — [GitHub](#)

Dec 2025

- Built and compared logistic regression, random forest, and neural-network models for heart-failure mortality prediction, achieving 81.3% validation accuracy and identifying the most influential clinical risk factors.

### UCLA ASA DataFest 2025

Apr 2025

- Analyzed 190K+ commercial-real-estate observations across 35 variables to identify geographic and industry-level business-office migration patterns and communicate business-focused findings.

## EXPERIENCE

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### MOORPARK COLLEGE ENGINEERING CLUB

Sep 2022 – May 2024

Lead Programmer | Board Member – Secretary

- Led a 3-person programming team in qualifier competitions and the 2024 VEX U Robotics World Championship.
- Coordinated software design, hardware-software integration, and debugging in time-constrained settings.
- Developed autonomous navigation and control algorithms in C++ using PID controllers, improving autonomous task success rates by over 2x.

## SKILLS

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**Programming:** Python, SQL, R, C++, MATLAB

**Libraries:** pandas, NumPy, scikit-learn, PyTorch, XGBoost, statsmodels, tidyverse, ggplot2

**Methods:** Regression, Classification, Causal Inference, Feature Engineering, Model Evaluation, Hyperparameter Tuning, EDA

**Tools:** Git, Jupyter, RStudio, Visual Studio, LaTeX, Quarto